

STAT 516: Basic Probability and its Applications

Lecture 2: Axioms of Probability. Calculating Probabilities. Inclusion-Exclusion Formula

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Axioms of Probability

- ▶ Let $\mathcal{S}$ be the sample space and $\mathcal{B}$ be the set of all events in $\mathcal{S}$ *whole sample space*
- ▶ *Non-negativity* For all $C \in \mathcal{B}$, $P(C) \geq 0$ *probabilities can't be negative numbers*
- ▶ *Normalization* $P(\mathcal{S}) = 1$ *probability of something happening in the defined space is 1.*
- ▶ *Countable Negativity* If $\{C_n\}$ is a sequence of events in $\mathcal{B}$ and $C_m \cap C_n = \emptyset$ for all $m \neq n$,

$$P\left(\bigcup_{n=1}^{\infty} C_n\right) = \sum_{n=1}^{\infty} P(C_n) \quad \text{countable additivity}$$

*If events are mutually exclusive, the probability of $\bigcup$ Union, one or the other, is added sum.
(disjoint, no overlap)*

Countable Additivity Axiom

- ▶ The last axiom above is called the **countable additivity axiom**
 - ▶ The following is implied by the countable additivity axiom...
 - ▶ Let $A_1 \supset A_2 \supset A_3 \supset \dots$ be an infinite family of subsets of a sample space $\mathcal{S}$ s.t. $A_n \downarrow A$
(nested) decreasing sequence of events
 $\bigcap_{n=1}^{\infty} A_n = A$
 - ▶ Then, $P(A_n) \rightarrow P(A)$ as $n \rightarrow \infty$
eventually, they shrink down to a 'limiting' event A.
- ✓ If the sets shrink toward nothing, their probabilities shrink toward 0.
 - ✓ If they shrink toward some remaining set, the probabilities converge to that set's probability.

Instead of directly proving for decreasing A ,
proof uses complement $B=A^c$ (Increasing sequences).

Countable Additivity Axiom



Continuity Probabilities

◦ Shrinking/expanding sequences have probabilities that converge.

- ▶ To prove, note if $B_i = A'_i$, we have $B_1 \subset B_2 \subset B_3 \cdots$ and $B_n \uparrow B$
Increasing Sequences are easier with unions.
- ▶ It is enough to show that $P(B_n) \rightarrow P(B)$ as $n \rightarrow \infty$
- ▶ For any fixed n , $B_n = \bigcup_{i=1}^n (B_i \setminus \underbrace{B_{i-1}}_{\text{previous}})$ *Limiting Complement Event*
- ▶ In the above, $B_0 = \emptyset$ and $B_i - B_{i-1} = B_i \cap B'_{i-1}$
- ▶ The needed result immediately follows...
- ▶ Note that if *finite additivity* and the above result are assumed as axioms, then countable additivity follows as a consequence

Simple and Compound Events

- ▶ An event is **simple** if it consists of exactly one outcome and **compound** otherwise.
- ▶ Example: consider recording the freeway exit (L or R) for each of the 3 vehicles at the end of the exit ramp. The eight possible outcomes form the sample space $\mathcal{B} = \{LLL, RLL, LRL, LLR, LRR, RLR, RRL, RRR\}$.
- ▶ Examples of simple events would be:
 - ▶ $E_1 = LLL$
 - ▶ $E_2 = LRL$

Examples of Compound Events

- ▶ Examples of ^{Multiple Outcomes} compound events would be:
 - ▶ Exactly one vehicle turns left $A = \{LRR, RLR, RRL\}$ ^{3 possible events}
 - ▶ All three vehicles turn in the same direction $B = \{LLL, RRR\}$ ^{2 possible events}

Disjoint Events I

$$\text{Axiom 3: } P(A \cup B) = P(A) + P(B)$$

- ▶ When A and B have no outcomes in common, they are said to be **mutually exclusive** or **disjoint** events.
- ▶ Example: when rolling a die, the event $A = \{2, 4, 6\}$ (evens) and $B = \{1, 3, 5\}$ (odds) are mutually exclusive $\text{Ex: } P(A) = 3/6 = 1/2$
 $A \cap B = \emptyset$
- ▶ Example: when pulling a single card from a standard deck of cards, events $A = \text{heart, diamond (red)}$ and $B = \text{spade, club (black)}$ are mutually exclusive.

Disjoint Events II

At least one of them occur.

↑ (Together, they cover the whole sample space)

- ▶ A collection of events $\{C_n\}$ is **exhaustive** if the union of these events is a sample space so that

$$\sum_{n=1}^{\infty} P(C_n) = 1$$

- ▶ A mutually ^①exclusive and ^②exhaustive collection of events forms a **partition** of $\mathcal{S}$
- ▶ Example: events A and B from the card example form a partition of the sample space

② $A \cup B = \mathcal{S}$
① $A \cap B = \emptyset$

A and B make up
a partition of sample space.
(part

EX): $A_1: \{1, 2, 3\}$
 $A_2: \{4, 5, 6\}$

Simplest Properties of Probability

Property 1 Complement Rule:

- For each $C \in \mathcal{B}$

Axiom 3: $P(C') = 1 - P(C)$



Property 2 Null Set Rule:

- The probability of the null set is zero:

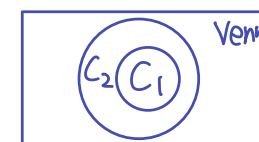
$$P(\emptyset) = 0$$

empty event never happens, so 0

Property 3 Monotonicity

- If $C_1 \subset C_2$ we have
smaller $\subset$ bigger subset can't have a higher probability

$$P(C_1) \leq P(C_2)$$



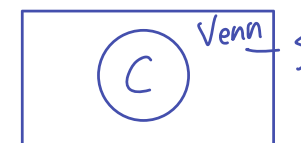
Property 4 Bounds

- For each event $C \in \mathcal{B}$

Axiom 1 $0 \leq P(C) \leq 1$,

Axiom 2 $P(S) = 1$

as $C \subset S$



$$P(C) \leq P(S) = 1$$

C and C' are disjoint events: $C \cup C' = S$ Total
 $1 = P(S) = P(C \cup C') = P(C) + P(C')$ Axiom 3
 $P(C') = 1 - P(C)$

Equilikely Case I

- ▶ Let $C_1, \dots, C_k$ be **mutually exclusive** and **exhaustive** subsets.
- ▶ Let each of $C_i, i = 1, \dots, k$ have the same probability $P(C_i) = \frac{1}{k}$
- ▶ For any $E = C_1 \cup C_2 \cup \dots \cup C_r$ with $r \leq k$

$$P(E) = \sum_{i=1}^r P(C_i) = \frac{r}{k}$$

This is the **classical definition of probability**:

1. Equally Likely Outcomes

- If all outcomes in the sample space are equally likely, then each has probability $1/k$.
- Example: A fair 6-sided die $\rightarrow$ each outcome has probability $1/6$.

2. Event Probability

- An event is just a subset of these outcomes.
- Probability of the event = number of favorable outcomes $\div$ total outcomes.
- Example: Event = "roll an even number."
 - Favorable = $\{2, 4, 6\}$ = 3 outcomes.
 - Probability = $3/6 = 1/2$.

Example I

Express everything
relative to the end car:

$$\text{Let } p_1 = p_5 = x$$

$$p_2 = p_4 = 2x$$

$$p_3 = 2 \cdot (2x) = 4x$$

$$\begin{aligned} p_1 + p_2 + p_3 + p_4 + p_5 \\ = x + 2x + 4x + 2x + x \\ = 10x \end{aligned}$$

- ▶ During off-peak hours a commuter train has five cars. Suppose a commuter is twice as likely to select the middle car #3 as to select either adjacent car (#2 or #4), and is twice as likely to select either adjacent car as to select either end car (#1 or #5).
- ▶ If the probability of selecting car i is p_i , we have $p_3 = 2p_2 = 2p_4$, and $p_2 = 2p_1 = 2p_5 = p_4$
- ▶ Therefore, $1 = \sum p_i = 10p_1$ and $p_1 = p_5 = 0.1$, $p_2 = p_4 = 0.2$ and $p_3 = 0.4$.
- ▶ The probability that one of the three middle cars is selected is, then, $p_2 + p_3 + p_4 = 0.8$.

Counting rules

- ▶ A **multiplication rule** or the *mn-rule*

- ▶ The number of **permutations** of k elements out of n is

✓ Order matters

Ex): "Arranging 3/9 ppl for CEO, COO, VP." P_9^3

$$P_k^n = n(n-1) \cdots (n-(k-1)) = \frac{n!}{(n-k)!}$$

choice 1 choice 2 ... fewer

arrange k

$n(n-1)(n-2) \cdots (n-(k-1)) = P_k^n$

k objects from a total of n

- ▶ The number of **combinations** of k elements out of n is

✗ Order doesn't matter

Ex): "Choose 3 members from 10 applicants." $\binom{10}{3}$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

every unique group of k item is counted

choose k

how many groups of size k you can choose from n items.

- ▶ The **binomial expansion**

Ex): $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
 $n=3$

count many ways to choose terms with terms $a^k b^{n-k}$ when expanding:

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$$

Probability set function for a finite set and loaded dice

- ▶ Let $\mathcal{S}$ consist of k distinct points (outcomes)
- ▶ Assign the probability of $\frac{1}{k}$ to each of these outcomes *Each outcome has an equal chance*
- ▶ For any event $C \in \mathcal{S}$ define

$$P(C) = \frac{\overset{\text{Total outcomes}}{\text{number of points in } C}}{k} = \sum_{x \in C} f(x)$$

with $f(x) = \frac{1}{k}$ for $x \in \mathcal{C}$

Counting rules

- without replacement* ▶ The number of ways of linearly arranging n distinct objects with order of arrangement matters is $n!$
- with replacement* ▶ The number of ways of choosing r objects from n distinct objects if the same object could be chosen repeatedly is n^r *not limited by # of choices.*
- ▶ The number of ways of distributing n distinct objects into k distinct categories when order in which the distributions are made is not important, n_i being the number of objects allocated to the i th category is
- $$\frac{n!}{n_1!n_2!\cdots n_k!} = \binom{n}{n_1} \binom{n-n_1}{n_2} \cdots \binom{n-n_1-n_2-\cdots-n_{k-1}}{n_k}$$

Example

without
replacement $n!$

Formula:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

- ▶ A carton of eggs has 12 eggs of which three are bad. We don't know that fact in advance
- Total ▶ We want to make a three-egg omelet; we can select three eggs in $\binom{12}{3} = 220$ ways
- Favorable ▶ Assume that three eggs are selected completely at random; the number of ways to select them out of 9 good eggs is $\binom{9}{3} = 84$
- ▶ The probability of the favorable outcome is $\frac{84}{220} = 0.38$

$$\begin{aligned}\frac{9!}{3!6!} &= \frac{9 \cdot 8 \cdot 7 \cdot \cancel{6!}}{3! \cdot \cancel{6!}} \\ &= \frac{9 \cdot 8 \cdot 7}{3 \cdot 2 \cdot 1} \\ &= \frac{504}{6} = 84\end{aligned}$$

Example

with
replacement n^r

- ▶ Six different cookies are distributed completely at random to six children. It is possible for a child to get more than one cookie
 *6 choices (1st), 6 choices (2nd)...
not limited by the number of cookies*
- ▶ This implies that there are $6^6 = 46,656$ possible outcomes (sample points).
- ▶ Let $A = \{\text{Each child receives exactly one cookie}\}$. There are $6! = 720$ ways of doing so and thus $P(A) = \frac{720}{46656} = 0.015$
 1.5% that each child gets exactly one cookie.
- ▶ Then, $P(A') = 1 - 0.015 = \underline{0.985}$

$$6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

6 choices
first child took one cookie, so 5 choices left...

Inclusion-Exclusion Formula

- ▶ $P(C_1 \cup C_2) = P(C_1) + P(C_2) - P(C_1 \cap C_2)$ *double counting*
- ▶ The general **inclusion-exclusion** formula

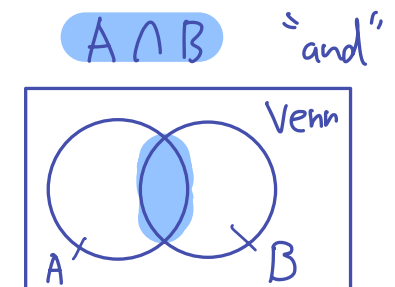
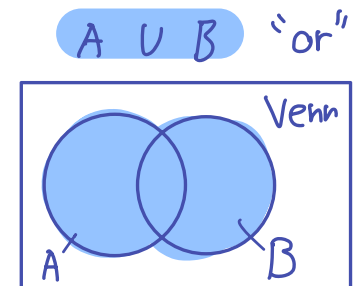
$$P(C_1 \cup C_2 \cup \dots \cup C_n) = \sum_{i=1}^n P(C_i) \textcircled{-} \sum_{1 \leq i < j \leq n} \textcircled{1} \text{ pairwise interactions } P(C_i \cap C_j) \\ \textcircled{+} \sum_{1 \leq i < j < k \leq n} \textcircled{2} \text{ three-way interactions } P(C_i \cap C_j \cap C_k) \\ \text{---} \dots + (-1)^{n+1} P(C_1 \cap C_2 \cap \dots \cap C_n)$$

$$P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3)$$

$$\boxed{-P(A_1 \cap A_2) - P(A_2 \cap A_3) - P(A_3 \cap A_1)} \quad \boxed{+P(A_1 \cap A_2 \cap A_3)}$$

$\textcircled{1}$ (alternating signs) $\textcircled{2}$

$$(-1)^{3+1} = (+1)$$



Example II

- ▶ In a residential suburb, 60% of all households get Internet service from a local cable company, 80% get television service from that company, and 50% get both services from that company
- ▶ The probability that a randomly selected household subscribes to at least one of these two services from the local company is

$$\begin{array}{c} \text{or} \\ P(C_1 \cup C_2) = P(C_1) + P(C_2) - P(C_1 \cap C_2) = 0.9 \\ \quad \quad \quad 0.6 \quad + \quad 0.8 \quad - \quad 0.5 \end{array}$$

Example II

- ▶ The probability that a household subscribes only to TV service is



$$P(C_1' \cap C_2) = P(C_1 \cup C_2) - P(C_1) = 0.9 - 0.6 = 0.3,$$

- ▶ The probability that a household subscribes only to Internet is

$$P(C_1 \cap C_2') = P(C_1 \cup C_2) - P(C_2) = 0.1,$$

- ▶ The probability that a household subscribes to exactly one of these services from the local company is *only TV or only internet*
Exactly one service

$$P(C_1' \cap C_2) + P(C_1 \cap C_2') = 0.1 + 0.3 = 0.4$$



Missing faces in dice rolls

- ▶ A fair die is rolled n times. What is the probability that at least one of six sides never shows up in these n rolls?

- ▶ Let A_i be the side that never shows up, $1 \leq i \leq 6$

- ✓▶ If all of 6^n outcomes are equally likely, $P(A_i) = \frac{5^n}{6^n}$, only 5 faces appear

only 4 faces appear $P(A_i \cap A_j) = \frac{4^n}{6^n}$, $P(A_i \cap A_j \cap A_k) = \frac{3^n}{6^n}$ etc only 3 faces appear

- ▶ Thus, the needed probability is **inclusion-exclusion** formula

$$p_n = \sum_{i=1}^6 \binom{6}{i} (-1)^{i+1} \left(\frac{6-i}{6}\right)^n$$

of sides to not show up 2 sides do not show up P(A_i ∩ A_j ∩ A_k)

one side is flipped, -1 side flip sign

- ▶ Manual calculation gives $p_{10} = 0.73$, $p_{12} = 0.56$, $p_{13} = 0.49$, $p_{15} = 0.36$

Ex: rolled 10x. $p_{10} = \binom{6}{1} \left(\frac{5}{6}\right)^{10} - \binom{6}{2} \left(\frac{4}{6}\right)^{10} + \binom{6}{3} \left(\frac{3}{6}\right)^{10} - \binom{6}{4} \left(\frac{2}{6}\right)^{10} + \binom{6}{5} \left(\frac{1}{6}\right)^{10} - \binom{6}{6} \left(\frac{0}{6}\right)^{10}$

- ▶ Note that it takes 13 rolls of a fair die to have better than 50% chance ($1 - p_n$) that each of the six sides shows up

Bonferroni bounds and inequality

- ▶ Denote $S_1 = \sum_{i=1}^n P(C_i)$, $S_2 = \sum_{1 \leq i < j \leq n} P(C_i \cap C_j)$, $S_3 = \sum_{1 \leq i < j < k \leq n} P(C_i \cap C_j \cap C_k)$ etc
- ▶ Then, the inclusion-exclusion formula becomes

$$P(\cup_{i=1}^n C_i) = S_1 - S_2 + S_3 - \dots$$
- ▶ Let $p_n = P(\cup_{i=1}^n C_i)$. The following series of bounds is true:

$$p_n \leq S_1; p_n \geq S_1 - S_2; p_n \leq S_1 - S_2 + S_3; \dots$$

Easier way

- ▶ Also, (**Bonferroni inequality**)

Lower Bound

$$P(\cap_{i=1}^n C_i) \geq 1 - \sum_{i=1}^n P(C'_i)$$

If each C_i has high individual probability,
 then their joint probability (that all occur) is at least:
 $1 - \sum P(\text{each one failing})$

Example

- $1 - 0.95 = 0.05$
Individual events are 95% likely
- ▶ Let $P(C_i) \geq 0.95$ for $i = 1, \dots, 10$. This means that $P(C'_i) \leq 0.05$ for each i
 - ▶ By Bonferroni inequality, $P(\cap_{i=1}^n C_i) \geq 1 - (10 * 0.05) = 0.5$
 - ▶ Therefore, despite individual probabilities of at least 95%, we can only be 50% sure that all 10 of events will occur
 - ▶ This is a rather crude lower bound - better options are available

◦ Use Bonferroni Bounds to approximate (bounds) for the union of probabilities when exact inclusion-exclusion is too long or messy to compute.

	With repetition	Without repetition
❖ Permutation: Order is important.	n^r	$n_{p_r} = \frac{n!}{(n-r)!}$
MS Excel Formula >>	PERMUTATIONA(n,r)	PERMUT(n,r)
❖ Combination: Order is NOT important.	$\frac{(r+n-1)!}{r!(n-1)!}$	$n_{c_r} = \frac{n!}{(n-r)! r!}$
MS Excel Formula >>	COMBINA(n,r)	COMBIN(n,r)

PERMUTATIONS / COMBINATIONS SUMMARY